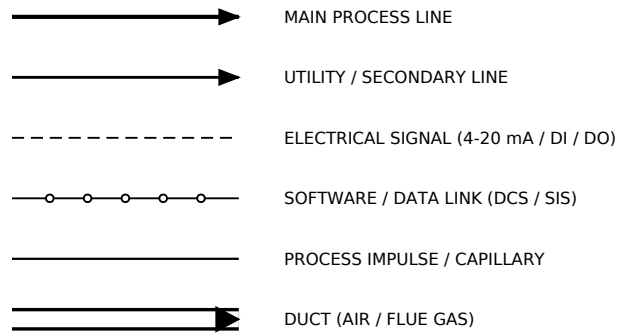
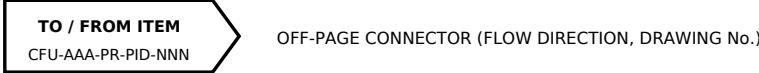
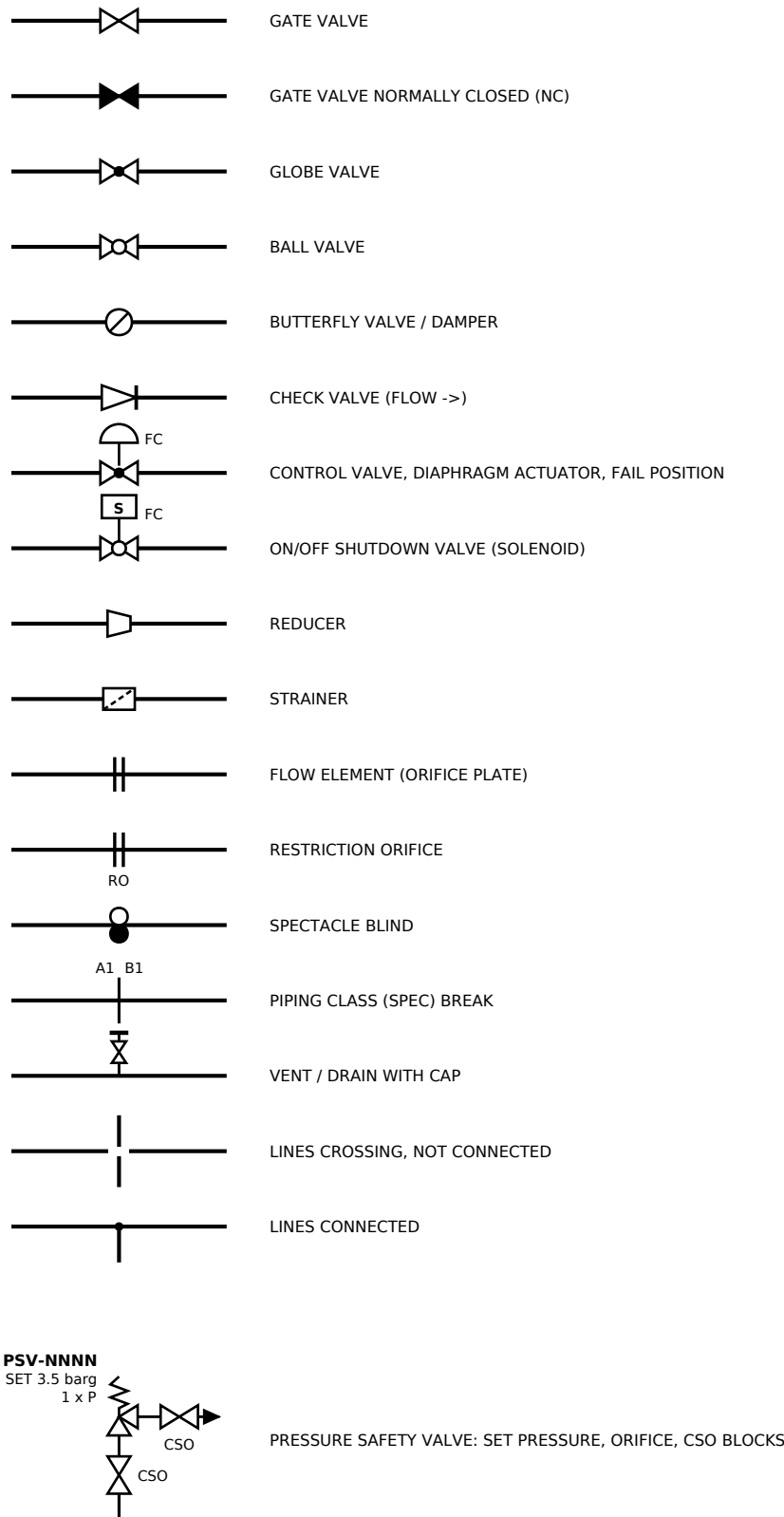


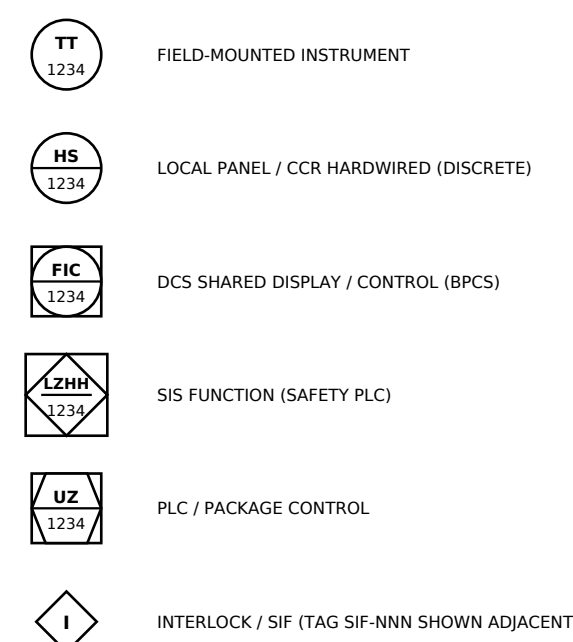
LINE SYMBOLS



VALVES & PIPING COMPONENTS



INSTRUMENT SYMBOLS (ISA-5.1)



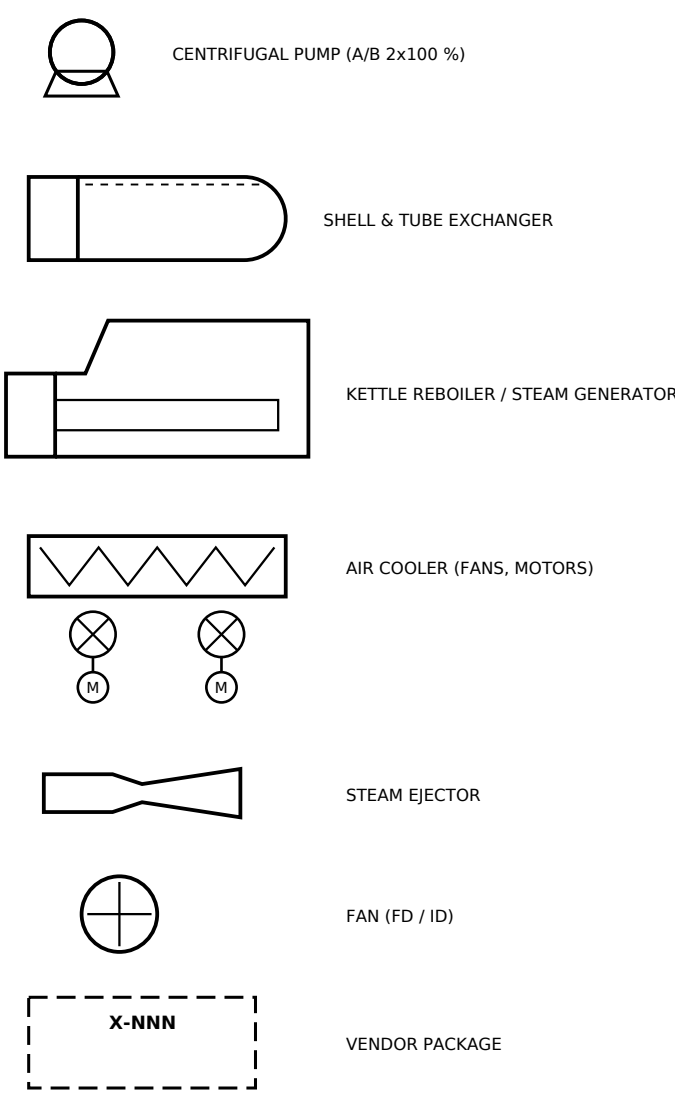
INSTRUMENT IDENTIFICATION LETTERS

A ANALYSIS (O2, pH) B BURNER / FLAME  
F FLOW H HAND  
L LEVEL P PRESSURE  
T TEMPERATURE X/Z ON-OFF / POSITION  
PD / TD DIFFERENTIAL FF FLOW RATIO  
Succeeding: E element, G glass, I indicate, C control,  
T transmit, V valve, Y relay/compute, S switch, Z (2nd) SIS,  
AH/AH/HI/ALL alarms; A/B/C suffix = redundant SIS

ABBREVIATIONS

FC / FO / FL FAIL CLOSED / OPEN / LAST  
CSO / CSC CAR-SEALED OPEN / CLOSED  
NC / NO NORMALLY CLOSED / OPEN  
NMF NORMALLY NO FLOW  
TPA / MPA / BPA TOP / MID / BOTTOM PUMP/AROUND  
SS STRIPPING STEAM  
DBB DOUBLE BLOCK & BLEED  
ROSOV / EIV REMOTE OPERATED SHUT-OFF VALVE  
COI / COT COIL OUTLET / INLET TEMP.  
RO RESTRICTION ORIFICE  
FA FLAME ARRESTOR  
OSBL OUTSIDE BATTERY LIMITS  
BMS BURNER MANAGEMENT SYSTEM  
2oo3 TWO-OUT-OF-THREE VOTING

EQUIPMENT SYMBOLS



LINE NUMBER

12"-P-100-002-B1-H

NPS (in) - FLUID - AREA - SEQUENCE - PIPING CLASS - INSULATION  
AREA: 100 CDU, 200 VDU, 900 UNIT UTILITIES

FLUID CODES

P Process hydrocarbon  
PG Process gas / vapour  
SW Sour water  
WW Wash water / brine  
LS LP steam  
HS HP steam  
CD Condensate  
BFW Boiler feed water

CWS Cooling water supply  
CWR Cooling water return  
FG Fuel gas  
FL Flare  
BD Blowdown / drain  
IA Instrument air  
N Nitrogen  
CH Chemical

INSULATION CODES

H HEAT CONSERVATION  
P PERSONNEL PROTECTION  
ST STEAM TRACED + INSULATED  
C COLD  
N NONE

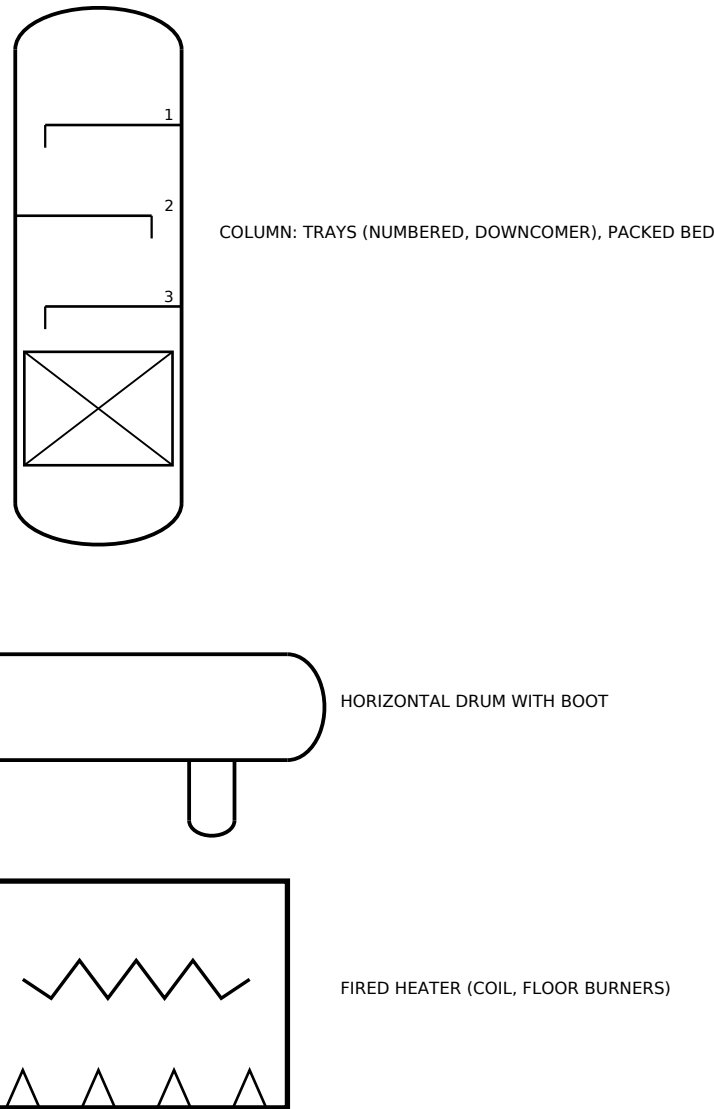
PIPING CLASSES (SUMMARY, SEE CFU-000-PI-SPC-001)

A1 150# Carbon steel ASTM A106 Gr.B  
A2 150# Carbon steel, HIC-resistant (NACE MR0103)  
B1 300# Carbon steel ASTM A106 Gr.B  
B2 300# SC-1/2Mo ASTM A335 P5  
B3 300# 9Cr-1Mo ASTM A335 P9  
C1 600# Carbon steel, killed ASTM A106 Gr.B  
A3 300# Carbon steel, HIC-resistant (NACE MR0103)  
S1 150# Carbon steel ASTM A106 Gr.B  
S2 600# 1.25Cr-1/2Mo ASTM A335 P11  
S3 300# Carbon steel ASTM A106 Gr.B  
F1 150# Carbon steel, killed, impact-tested (A33)  
U1 150# Carbon steel (cement-lined for CW >= 12")  
U2 150# SS304 (A) / galvanised CS (N2)  
DRAWING INDEX-335 P5

Hydrocarbon <= 230 C, general  
Sour water, OH vapour/condensate, wash water/brine  
Hydrocarbon <= 260 C, pump discharges  
Sulfidic hydrocarbon 260-400 C (incl crude, AR, BPA, HvGO, VR)  
Heater outlets / transfer lines (H-101, H-201) and > 400 C  
LPG / stabiliser, BFW > 300#  
Wash water / brine / sour water above 150# rating (desalter pr  
LP / HP steam, condensate  
HP steam 41 barg / 400 C  
MP steam 10.3 barg / 250-290 C design (exceeds S1)  
Flare / relief headers, -29 to 350 C  
Cooling water, utility water, LP BFW  
Instrument air, nitrogen  
Vacuum vapour lines (C-201 OH); transfer line > 400 C -> B3

LEGEND, SYMBOLS & ABBREVIATIONS  
CRUDE CHARGE PUMPS & COLD PREHEAT TRAIN  
ELECTROSTATIC DESALTERS D-101A/B & BOOSTER PUMPS  
HOT PREHEAT TRAIN E-106 TO E-111 & E-113  
ATMOSPHERIC HEATER H-101 - PROCESS COILS  
ATMOSPHERIC HEATER H-101 - FIRING & AIR SYSTEM  
ATMOSPHERIC COLUMN C-101 - FLASH ZONE & BOTTOMS  
ATMOSPHERIC COLUMN C-101 - UPPER SECTION & PUMP/AROUNDS  
SIDE STRIPPERS C-102/103/104 & PRODUCT RUNDOWN  
C-101 OVERHEAD SYSTEM & REFLUX DRUM D-102  
NAPHTHA STABILISER C-105  
NAPHTHA SPLITTER C-106  
VACUUM HEATER H-201  
VACUUM COLUMN C-201  
VACUUM PUMP/AROUNDS & PRODUCTS  
VACUUM EJECTOR SYSTEM & HOTWELL D-201  
FUEL GAS, FLARE KO & UTILITY HEADERS

EQUIPMENT DATA BOX

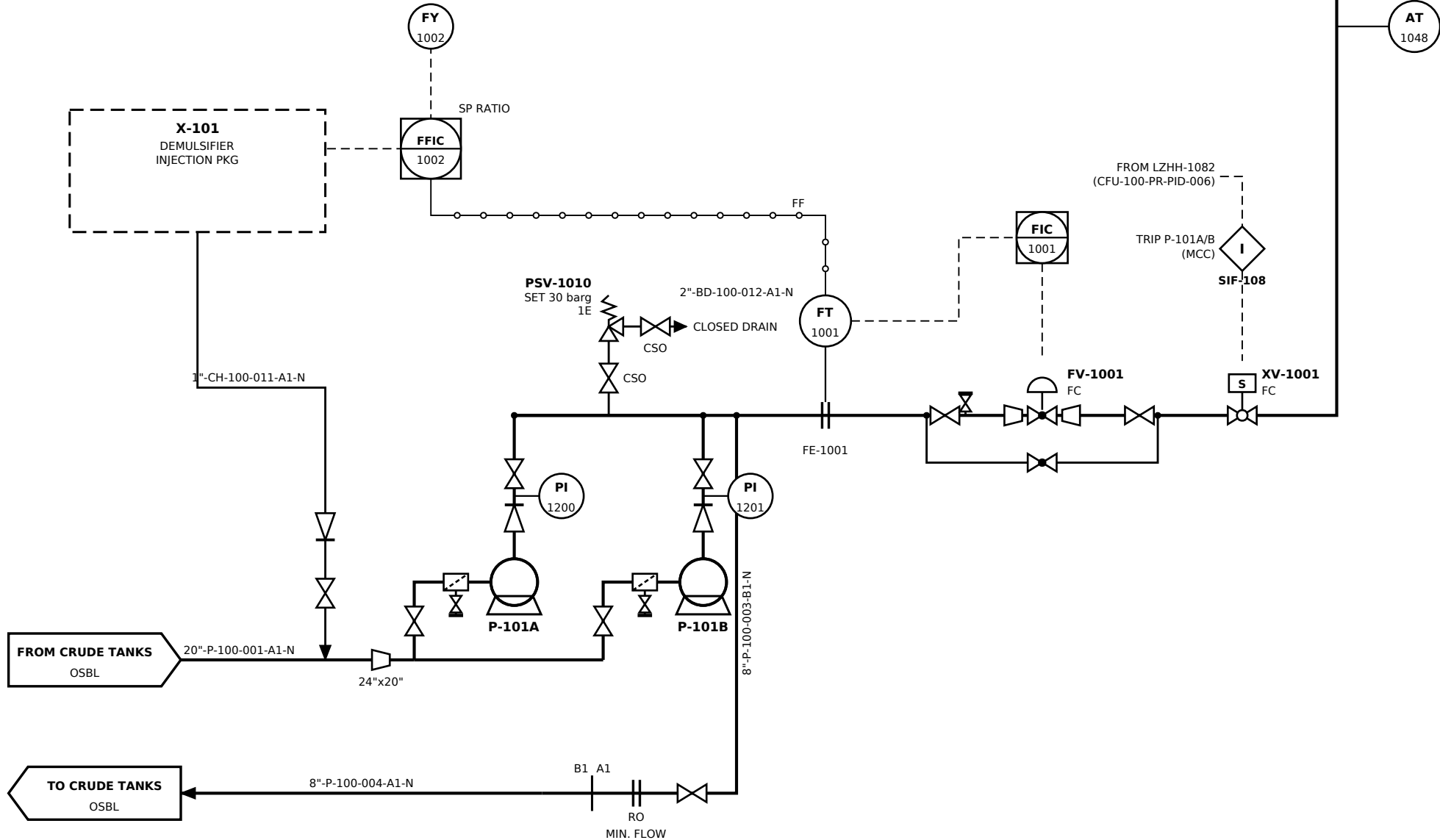
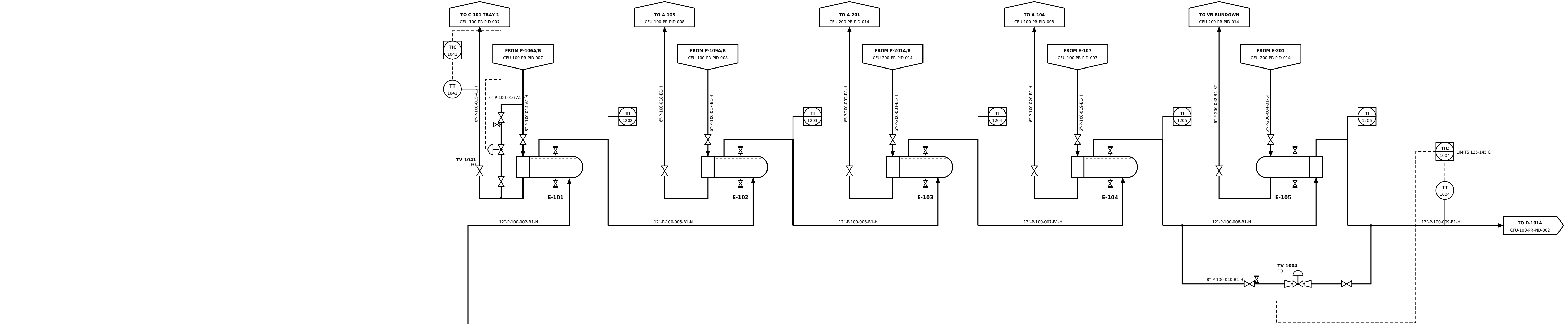


NOTES

- Legend applies to all P&IDs CFU-xxx-PR-PID-001 to -016.
- F&G devices (GD gas / H2S, BE flame, HS manual call points) are listed in instrument index CFU-000-IC-IDX-001; F&G layout drawing by others.
- DCS computing blocks (FY / PY / FFY) per control scheme drawings CFU-xxx-IC-CSD; listed in index.

| REV                                                                         | DATE       | DESCRIPTION              |            |
|-----------------------------------------------------------------------------|------------|--------------------------|------------|
| A                                                                           | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM<br>LEGEND, SYMBOLS & ABBREVIATIONS              |            |                          |            |
| DISCIPLINE                                                                  | STAGE      | SCALE                    | SHEET      |
| PROCESS                                                                     | FEED       | NTS                      | 1 OF 17    |
| DRAWN                                                                       | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                                                 | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                                                              |            |                          | REV        |
| CFU-000-PR-PID-000                                                          |            |                          | A          |

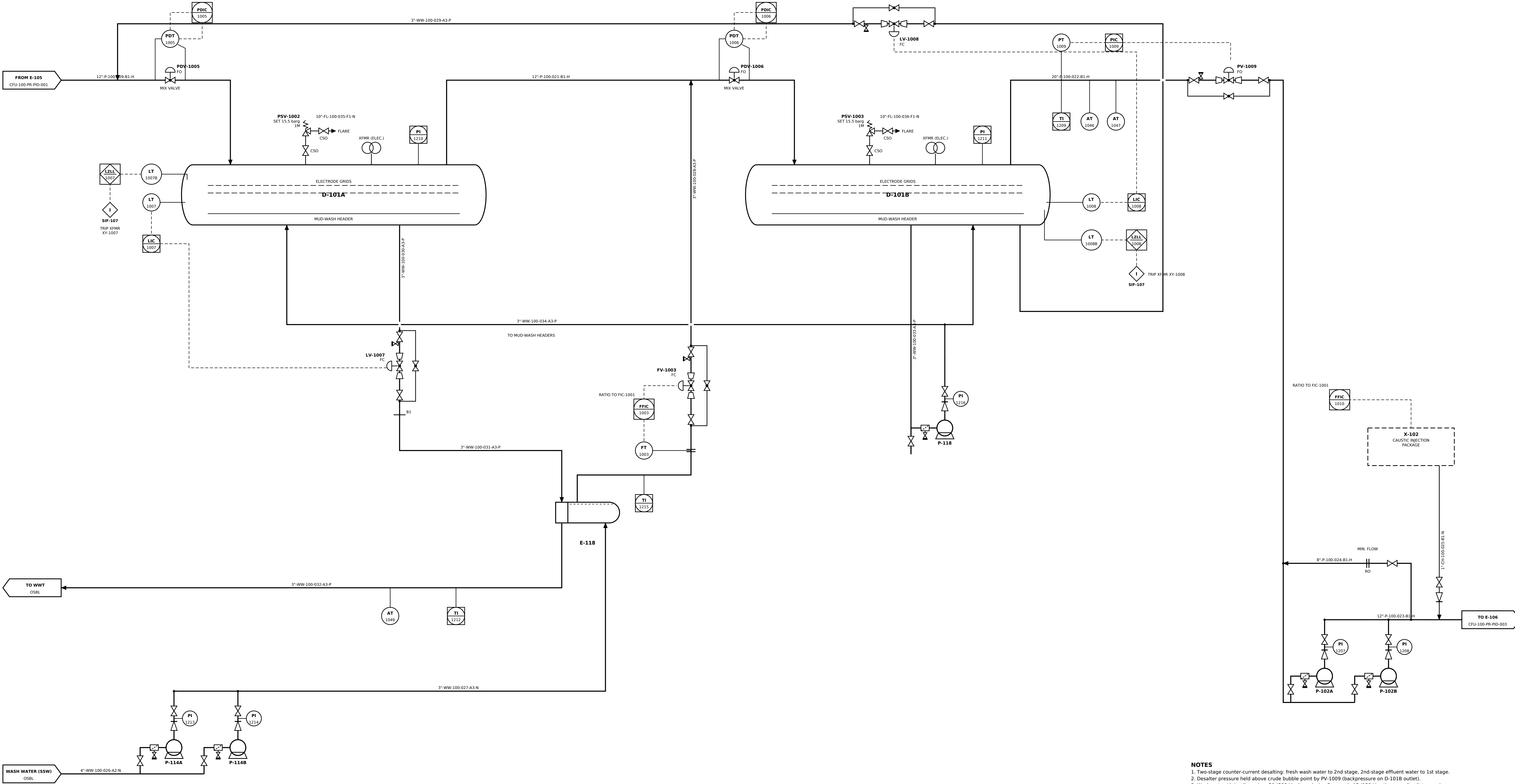
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| <b>P-101A/B</b><br>Crude charge<br>RATED 730 m3/h @ 261 m, dP 22 bar<br>MOTOR 710 kW, API 682, Flm 11/22<br>DES. T 60 C; MAT'L S-6 (CS/12Cr) | <b>E-101</b><br>Crude / Top pumparound<br>359 m2 total; 15 x 1P shells<br>DESIGN 35 / 20 barg / 185 C; CA 3 mm<br>DUTY 8.09 MW<br>SHELL: Crude / TUBE: Top pumparound<br>MAT'L CS shell / CS tubes | <b>E-102</b><br>Crude / Kerosene product<br>438 m2 total; 15 x 1P shells<br>DESIGN 35 / 20 barg / 240 C; CA 3 mm<br>DUTY 6.55 MW<br>SHELL: Crude / TUBE: Kerosene product<br>MAT'L CS shell / CS tubes | <b>E-103</b><br>Crude / LVGO pumparound/product<br>804 m2 total; 25 x 1P shells<br>DESIGN 35 / 20 barg / 255 C; CA 3 mm<br>DUTY 11.45 MW<br>SHELL: Crude / TUBE: LVGO pumparound/product<br>MAT'L CS shell / CS tubes | <b>E-104</b><br>Crude / Diesel product<br>378 m2 total; 15 x 1P shells<br>DESIGN 35 / 20 barg / 235 C; CA 3 mm<br>DUTY 4.44 MW<br>SHELL: Crude / TUBE: Diesel product<br>MAT'L CS shell / CS tubes | <b>E-105</b><br>Crude / Vacuum residue<br>431 m2 total; 15 x 1P shells<br>DESIGN 35 / 20 barg / 285 C; CA 3 mm<br>DUTY 5.84 MW<br>SHELL: Vacuum residue / TUBE: Crude<br>MAT'L CS shell / 5Cr-1/2Mo tubes | <b>X-101</b><br>Demulsifier injection package<br>Tank 2 m3 + 2 x 100 % metering pumps<br>MOTOR 0.75 kW<br>MAT'L SS316 |
|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|



- NOTES**
1. Cold-train crude-side design 35 barg (E-101, E-105) >= P-101 shut-off; PSV-1010 (30 barg) thermal / blocked-in.
  2. Exchanger hot-side connections shown at channel / shell nozzles; TEMA type & shells per data sheets.
  3. TV-1041 / TV-1004 bypass control valves have block valves only (exchanger provides bypass path).
  4. Demulsifier dosed at P-101 suction, ratio to FIC-1001 (FFIC-1002) - X-101 vendor package.
  5. Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  6. All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  7. Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

|                                            |            |                          |            |
|--------------------------------------------|------------|--------------------------|------------|
| REV                                        | DATE       | DESCRIPTION              |            |
| A                                          | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST              |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |            |
| CRUDE CHARGE PUMPS & COLD PREHEAT TRAIN    |            |                          |            |
| DISCIPLINE                                 | STAGE      | SCALE                    | SHEET      |
| PROCESS                                    | FEED       | NTS                      | 2 OF 17    |
| DRAWN                                      | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                             |            |                          | REV        |
| CFU-100-PR-PID-001                         |            |                          | A          |

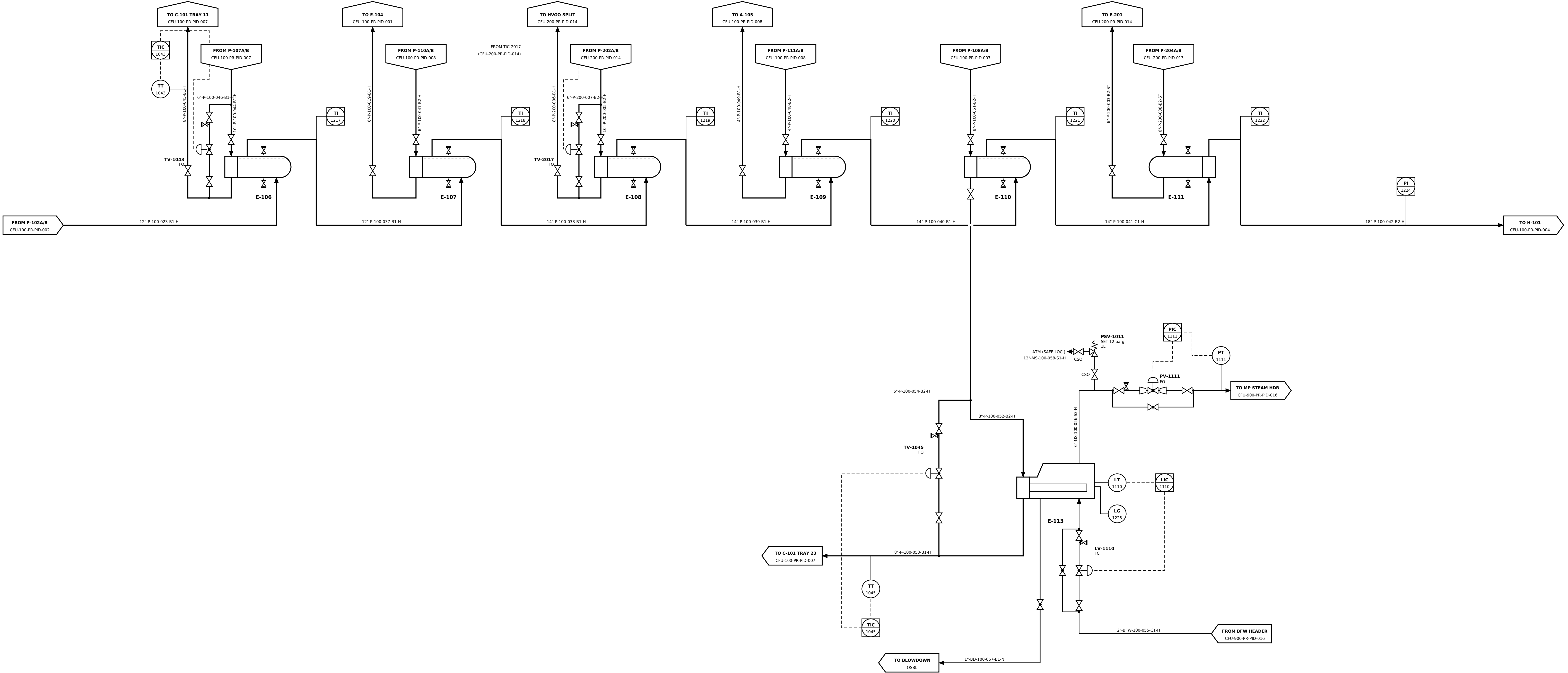
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| <b>D-101A</b><br>Electrostatic desalter, 1st stage<br>ID 3.8 m x 30.0 m TTT, ACDC dual-polarity grids<br>DESIGN 17.0 barg / 165 C, CA 3 mm<br>MAT'L CS | <b>D-101B</b><br>Electrostatic desalter, 2nd stage<br>ID 3.8 m x 30.0 m TTT, ACDC dual-polarity grids<br>DESIGN 17.0 barg / 165 C, CA 3 mm<br>MAT'L CS | <b>E-118</b><br>Desalter wash water / brine<br>84 m2, 1 shell(s)<br>DESIGN 20 / 20 barg / 160 C, CA 3 mm<br>DUTY 1.62 MW<br>SHELL: Wash water / TUBE: Brine<br>MAT'L CS shell / duplex 2205 tubes | <b>P-102A/B</b><br>Desalted crude booster<br>RATED 805 m3/h @ 263 m, dP 20 bar<br>MOTOR 713 kW, API 682, Plan 11052<br>DES. T 165 C, MAT'L S-6 (CS/12Cr) | <b>P-114A/B</b><br>Desalter wash water<br>RATED 37 m3/h @ 144 m, dP 14 bar<br>MOTOR 30 kW, API 682, Plan 11052<br>DES. T 80 C, MAT'L S-6 (CS/12Cr) | <b>P-118</b><br>Desalter mud-wash / recycle<br>RATED 19 m3/h @ 64 m, dP 6 bar<br>MOTOR 7.5 kW, API 682, Plan 11052<br>DES. T 150 C, MAT'L S-6 (CS/12Cr) | <b>X-102</b><br>Caustic injection package (desalted crude)<br>Tank 2 m3 + 2 x 100 % metering pumps<br>MOTOR 0.75 kW<br>MAT'L SS316 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|



- NOTES**
- Two-stage counter-current desalting; fresh wash water to 2nd stage, 2nd-stage effluent water to 1st stage.
  - Desalter pressure held above crude bubble point by PV-1009 (backpressure on D-101B outlet).
  - Wash water / brine lines class B1 (300#) where design P exceeds A2 (150#) rating - see line list remarks.
  - Transformer trip on low-low interface (SIF-107) by Electrical (XY-1007 / XY-1008 to feeder CB).
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                                                             | DATE       | DESCRIPTION              |            |
|---------------------------------------------------------------------------------|------------|--------------------------|------------|
| A                                                                               | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT     |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM<br>ELECTROSTATIC DESALTERS D-101A/B & BOOSTER PUMPS |            |                          |            |
| DISCIPLINE                                                                      | STAGE      | SCALE                    | SHEET      |
| PROCESS                                                                         | FEED       | NTS                      | 3 OF 17    |
| DRAWN                                                                           | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                                                     | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                                                                  |            |                          | REV        |
| CFU-100-PR-PID-002                                                              |            |                          | A          |

|                                                                                                                                                                                                                               |                                                                                                                                                                                                          |                                                                                                                                                                                                                                |                                                                                                                                                                                                      |                                                                                                                                                                                                                                    |                                                                                                                                                                                                             |                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>E-106</b><br>Crude / Middle (kero) pumparound<br>1001 m2 total; 25 x 1P shells<br>DESIGN 45 / 20 barg / 270 C; CA 3 mm<br>DUTY 5.83 MW<br>SHELL: Crude / TUBE: Middle (kero) pumparound<br>MATL:CS shell / 5Cr-1/2Mo tubes | <b>E-107</b><br>Crude / Diesel product<br>324 m2 total; 15 x 1P shells<br>DESIGN 45 / 20 barg / 325 C; CA 6 mm<br>DUTY 5.83 MW<br>SHELL: Crude / TUBE: Diesel product<br>MATL:CS shell / 5Cr-1/2Mo tubes | <b>E-108</b><br>Crude / HVGO pumparound/product<br>2196 m2 total; 25 x 2P shells<br>DESIGN 45 / 20 barg / 345 C; CA 6 mm<br>DUTY 21.14 MW<br>SHELL: Crude / TUBE: HVGO pumparound/product<br>MATL:1.25Cr shell / 9Cr-1Mo tubes | <b>E-109</b><br>Crude / AGO product<br>263 m2 total; 15 x 1P shells<br>DESIGN 45 / 20 barg / 345 C; CA 6 mm<br>DUTY 3.11 MW<br>SHELL: Crude / TUBE: AGO product<br>MATL:1.25Cr shell / 9Cr-1Mo tubes | <b>E-110</b><br>Crude / Bottom (diesel) pumparound<br>886 m2 total; 15 x 2P shells<br>DESIGN 45 / 20 barg / 345 C; CA 6 mm<br>DUTY 7.58 MW<br>SHELL: Crude / TUBE: Bottom (diesel) pumparound<br>MATL:1.25Cr shell / 9Cr-1Mo tubes | <b>E-111</b><br>Crude / Vacuum residue<br>1094 m2 total; 15 x 2P shells<br>DESIGN 45 / 20 barg / 390 C; CA 6 mm<br>DUTY 7.58 MW<br>SHELL: Vacuum residue / TUBE: Crude<br>MATL:1.25Cr shell / 9Cr-1Mo tubes | <b>E-113</b><br>BPA / MP steam generator (kettle)<br>223 m2; 1 shell(s)<br>DESIGN 13 / 20 barg / 290 C; CA 3 mm<br>DUTY 4.91 MW<br>SHELL: BFW / MP steam / TUBE: Bottom pumparound<br>MATL:CS |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



- NOTES**
- Hot-train crude-side design 45 barg (E-106, E-111) >= P-102 shut-off (D-1018 design + 1.2 x dPI).
  - E-113 kettle MP steam generator: 3-element level control simplified to LIC-1110 at FEED.
  - TV-2017 controlled from TIC-2017 (HVGO pumparound return temperature) on CFU-200-PR-PID-014.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                                                         | DATE       | DESCRIPTION              |            |
|-----------------------------------------------------------------------------|------------|--------------------------|------------|
| A                                                                           | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM<br>HOT PREHEAT TRAIN E-106 TO E-111 & E-113     |            |                          |            |
| DISCIPLINE                                                                  |            | STAGE                    | SHEET      |
| PROCESS                                                                     |            | FEED                     | 4 OF 17    |
| SCALE                                                                       |            | NTS                      |            |
| DRAWN                                                                       | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                                                 | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                                                              |            |                          | REV        |
| CFU-100-PR-PID-003                                                          |            |                          | A          |

**H-101**  
Atmospheric crude charge heater  
60.7 MW abs. / 67.4 MW fired; 2-cell cabin, 8 passes, 144 rad. tubes 6" x 18.3 m  
DESIGN 38.5 barg / 545 C; CA 3 mm  
DUTY 60.68 MW  
MAT'L Tubes A335 P9 (9Cr-1Mo); convection P5 / CS studded

LP STEAM HEADER  
CFU-900-PR-PID-016

FROM E-111 (CIT)  
CFU-100-PR-PID-003

BFW HEADER  
CFU-900-PR-PID-016  
1" BFW-100-072-016-H

TV-1024  
FC

TIC  
1024

350 C  
TT  
1024

PI  
1227

STRIPPING STEAM HDR  
CFU-100-PR-PID-006

TT  
1020A-C

2003  
SIF-106

TO C-101 FLASH ZONE  
CFU-100-PR-PID-006

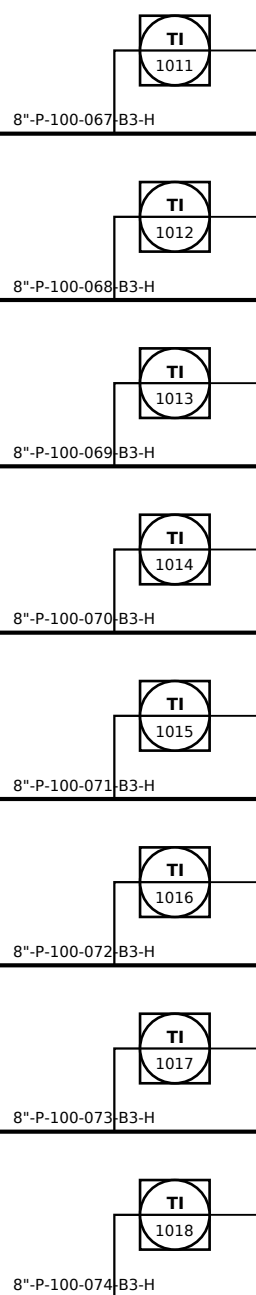
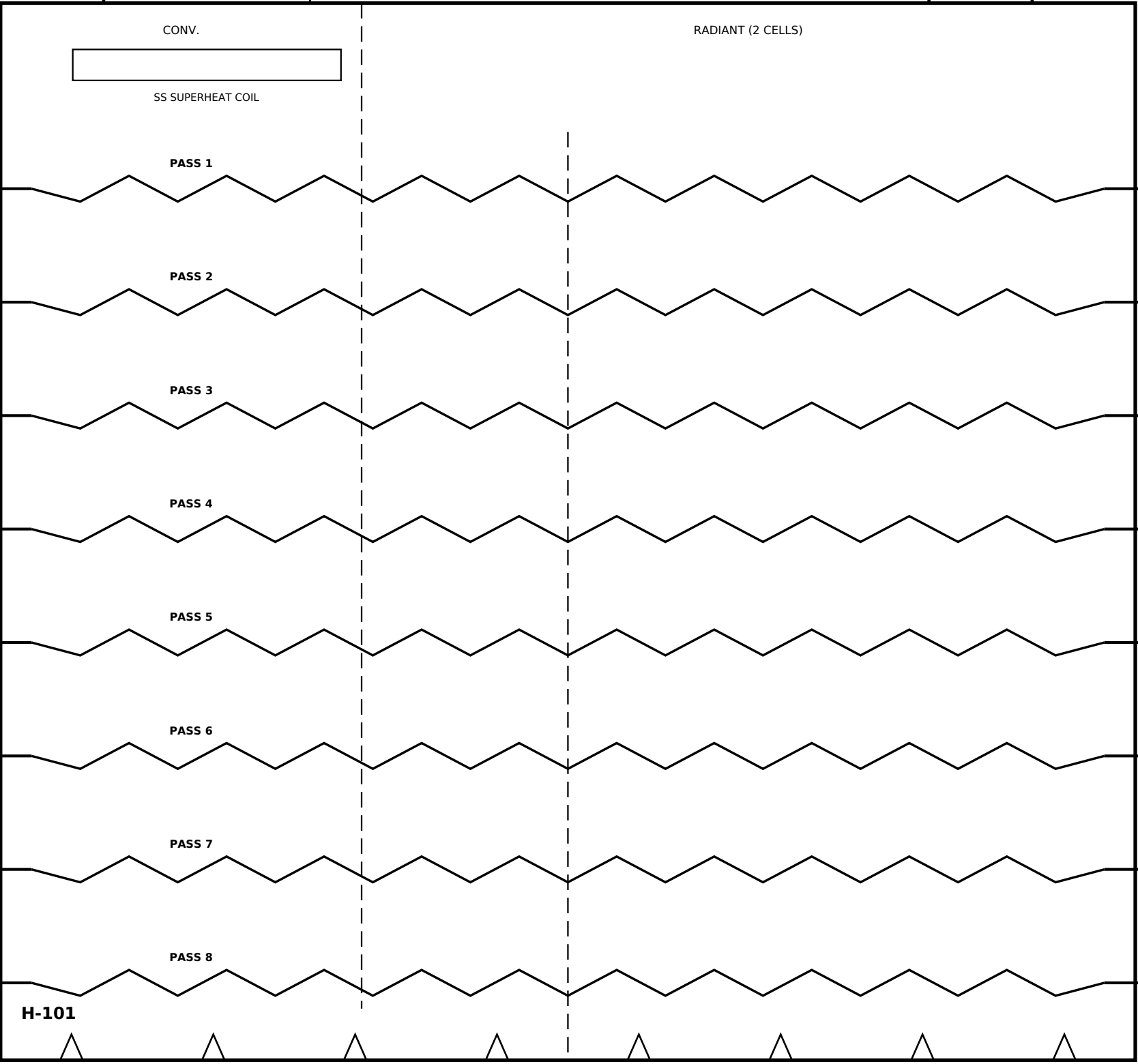
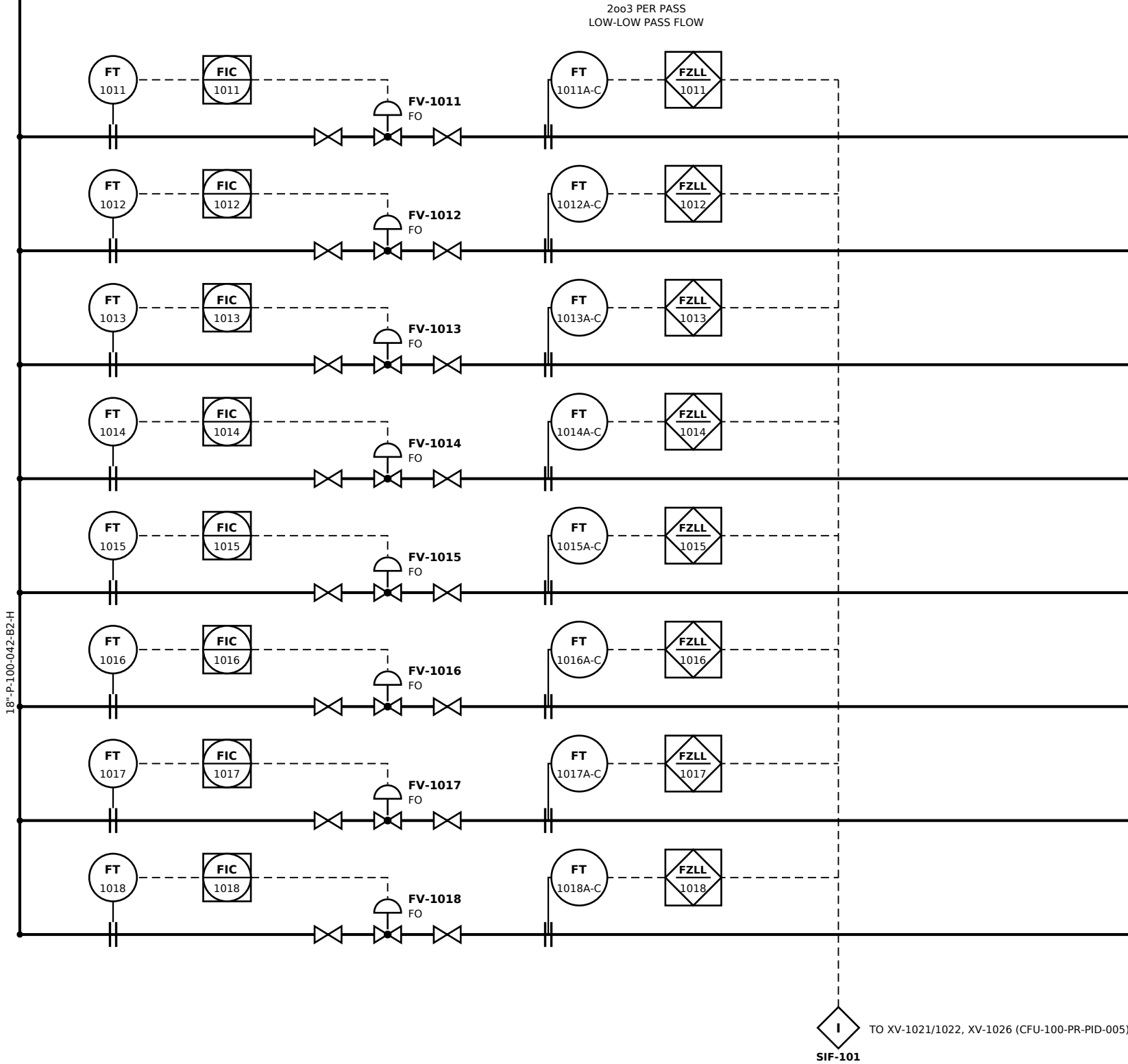
TT  
1020

TIC  
1020

SP CASCADE TO PIC-1021  
(CFU-100-PR-PID-005); FF FROM FIC-1001

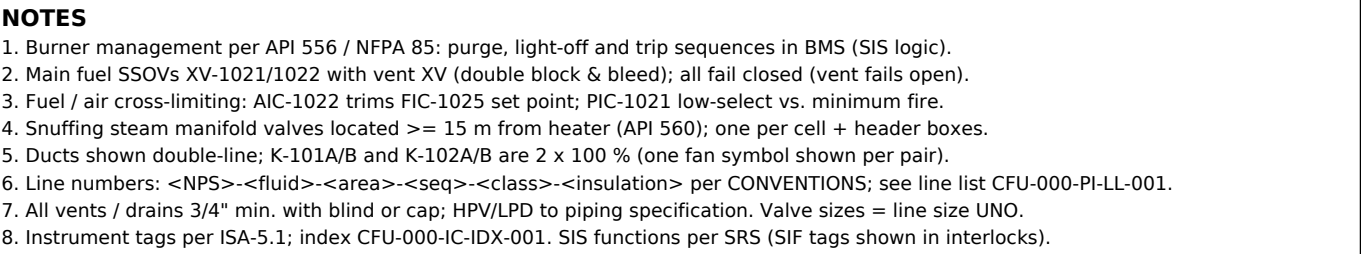
TDIC  
1019

BIAS TO FIC-1011-1018  
(TOTAL FLOW HELD)



- NOTES**
- Passes shown schematically (convection + radiant in series per pass); 8 identical passes.
  - Tube-skin thermocouples: 4 per pass (TI-1101..1132) - shown on heater vendor P&ID, alarmed in DCS.
  - Pass flow SIS transmitters FT-10xxA/B/C (2003 per pass) independent from BPCS FT-10xx (SIF-101).
  - Decoking (steam-air) connections and spool pieces at inlet / outlet manifolds by heater vendor.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                        | DATE       | DESCRIPTION              |
|--------------------------------------------|------------|--------------------------|
| A                                          | 2026-10-02 | Issued for review (FEED) |
| GENERIC REFINERY - GULF COAST              |            |                          |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |
| ATMOSPHERIC HEATER H-101 - PROCESS COILS   |            |                          |
| DISCIPLINE                                 | STAGE      | SCALE                    |
| PROCESS                                    | FEED       | NTS                      |
|                                            |            | SHEET                    |
|                                            |            | 5 OF 17                  |
| DRAWN                                      | CHECKED    | APPROVED                 |
| CLAUDE CODE                                | -          | -                        |
| DATE                                       |            | 2026-10-02               |
| DRAWING NUMBER                             |            | REV                      |
| CFU-100-PR-PID-004                         |            | A                        |

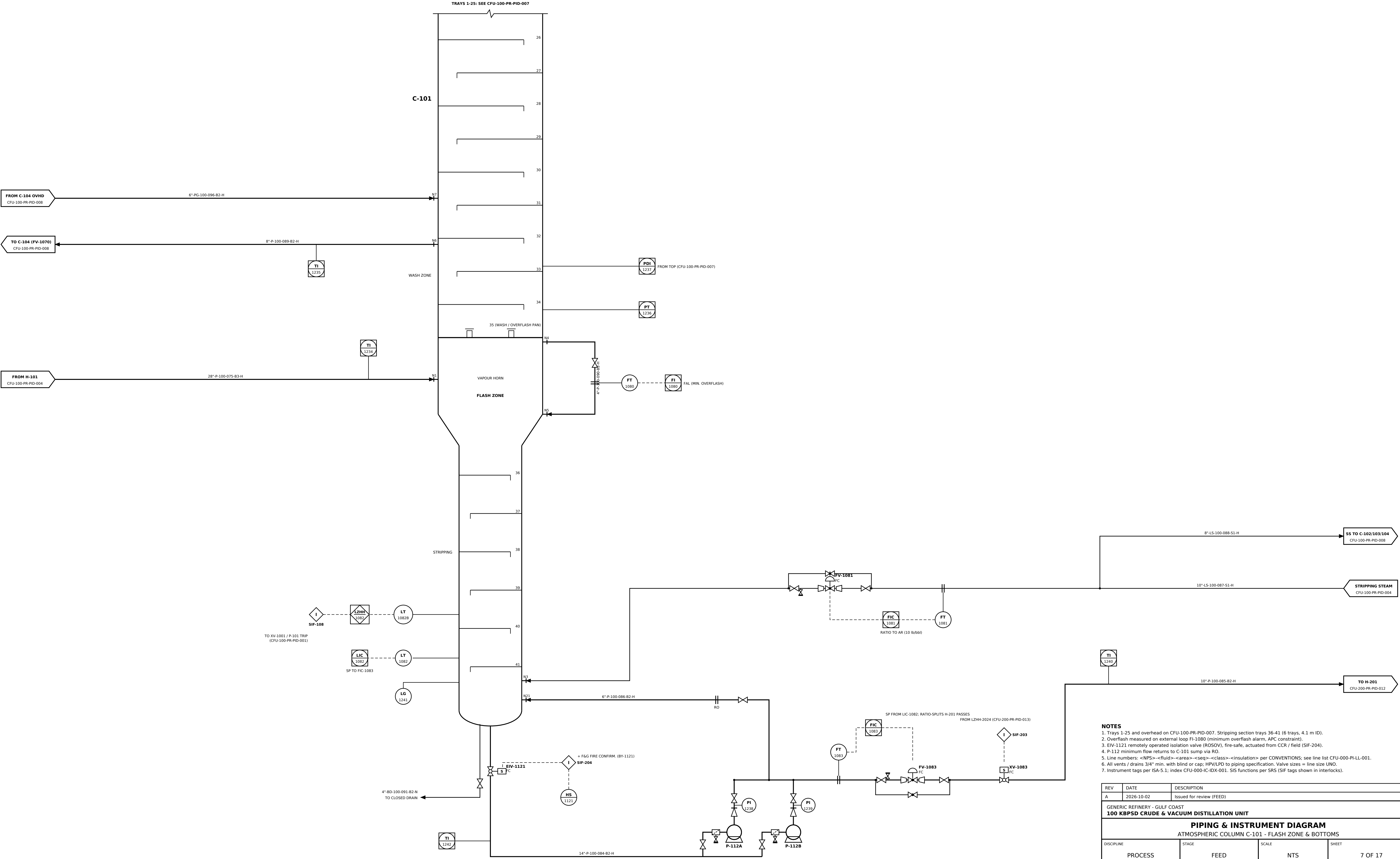


### PIPING & INSTRUMENT DIAGRAM

ATMOSPHERIC HEATER H-101 - FIRING & AIR SYSTEM

**C-101**  
Atmospheric crude fractionator  
ID 6.9 m (top/main) / 3.5 m (stripping) x 37.0 m T/T  
DESIGN 3.5 barg / 390 C, CA 3 mm  
41 valve trays (2/4-pass), 410S; Monel-lined top 5 trays  
MATL CS + 410S clad below tray 10; Monel 400 lined top head & trays 1-5

**P-112A/B**  
Atm. residue / vacuum heater charge  
RATED 398 m3/h @ 244 m, dP 17 bar  
MOTOR 315 kW, API 602 Plan 538  
DES. T 380 C; MATL A-8 (316)



**NOTES**  
1. Trays 1-25 and overhead on CFU-100-PR-PID-007. Stripping section trays 36-41 (6 trays, 4.1 m ID).  
2. Overflash measured on external loop FI-1080 (minimum overflash alarm, APC constraint).  
3. EIV-1121, remotely operated isolation valve (ROSOV), fire-safe, actuated from CCR / field (SIF-204).  
4. P-112 minimum flow returns to C-101 sump via RO.  
5. Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.  
6. All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.  
7. Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV | DATE       | DESCRIPTION              |
|-----|------------|--------------------------|
| A   | 2026-10-02 | Issued for review (FEED) |

GENERIC REFINERY - GULF COAST

**100 KBPSD CRUDE & VACUUM DISTILLATION UNIT**

**PIPING & INSTRUMENT DIAGRAM**

ATMOSPHERIC COLUMN C-101 - FLASH ZONE & BOTTOMS

| DISCIPLINE | STAGE | SCALE | SHEET   |
|------------|-------|-------|---------|
| PROCESS    | FEED  | NTS   | 7 OF 17 |

| DRAWN       | CHECKED | APPROVED | DATE       |
|-------------|---------|----------|------------|
| CLAUDE CODE | -       | -        | 2026-10-02 |

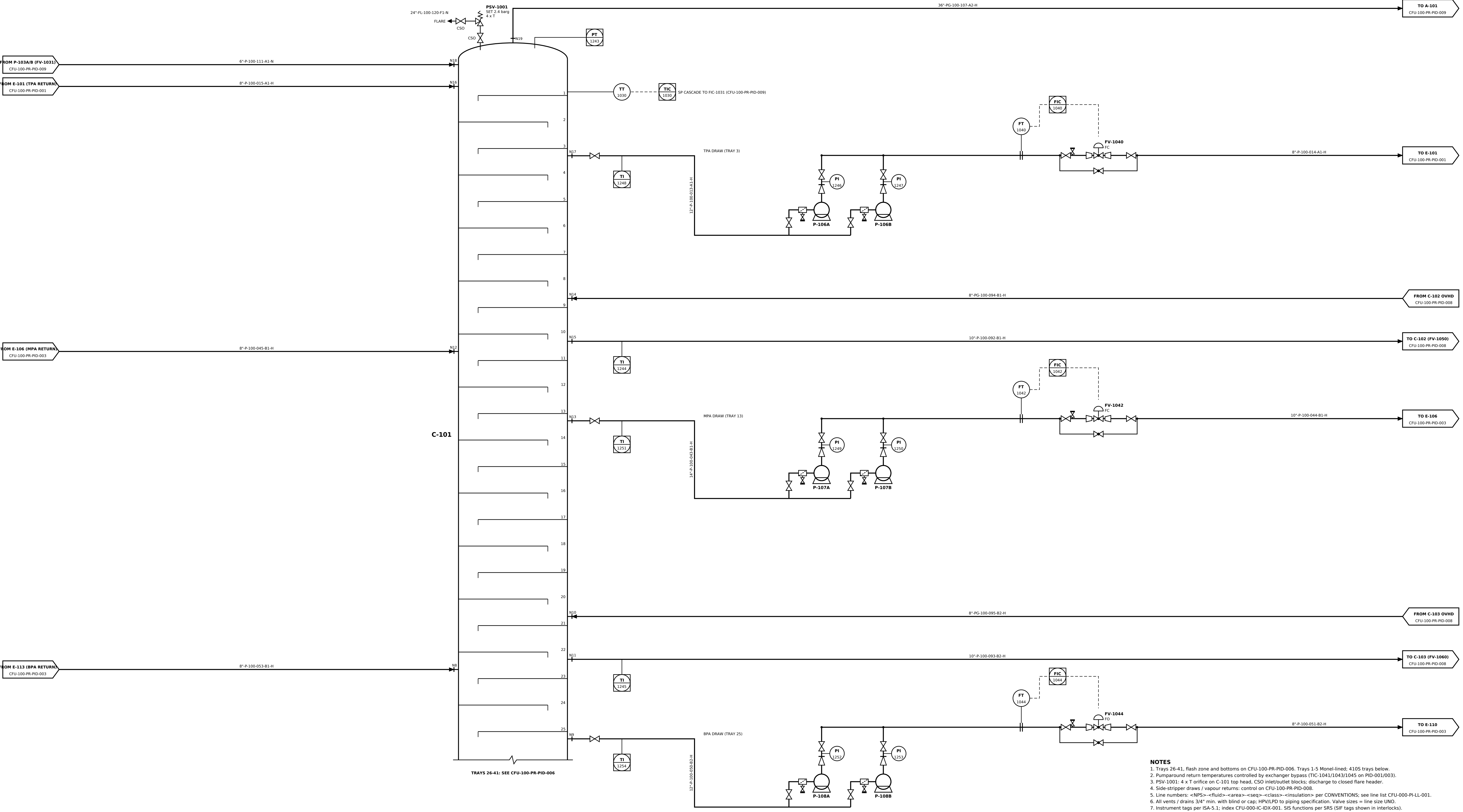
| DRAWING NUMBER            | REV      |
|---------------------------|----------|
| <b>CFU-100-PR-PID-006</b> | <b>A</b> |

**C-101**  
Atmospheric crude fractionator  
ID 6.9 m (top/main) / 3.5 m (stripping) x 37.0 m T/T  
DESIGN 3.5 barg / 300 C; CA 3 mm  
41 valve trays (2/4-pass), 4105; Monel-lined top 5 trays  
MAT'L CS + 4105 clad below tray 10; Monel 400 lined top head & trays  
1-5

**P-106A/B**  
TPA pumparound  
RATED 293 m<sup>3</sup>/h @ 130 m, dP 8 bar  
MOTOR 1104 kW, API 682 Plus 1152  
DES. T 185 C; MAT'L S-6 (CS/12Cr)

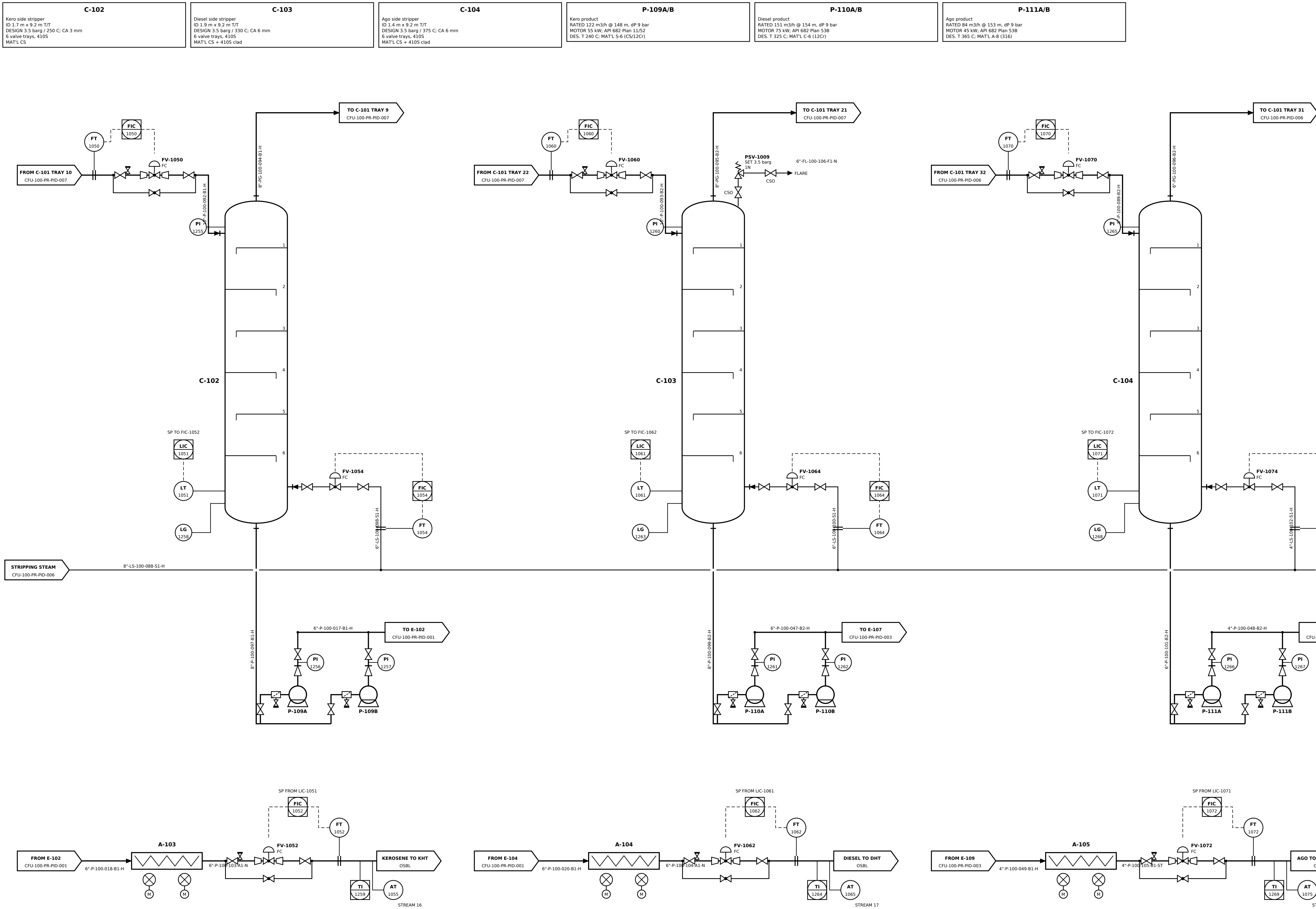
**P-107A/B**  
MPA pumparound  
RATED 387 m<sup>3</sup>/h @ 138 m, dP 8 bar  
MOTOR 1364 kW, API 682 Plus 1152  
DES. T 270 C; MAT'L C-6 (12Cr)

**P-108A/B**  
BPA pumparound  
RATED 318 m<sup>3</sup>/h @ 142 m, dP 8 bar  
MOTOR 1324 kW, API 682 Plus 538  
DES. T 345 C; MAT'L A-8 (316)



- NOTES**
- Trays 26-41, flash zone and bottoms on CFU-100-PR-PID-006. Trays 1-5 Monel-lined; 4105 trays below.
  - Pumparound return temperatures controlled by exchanger bypass (TIC-1041/1043/1045 on PID-001/003).
  - PSV-1001: 4 x T orifice on C-101 top head. CSO inlet/outlet blocks: discharge to closed flare header.
  - Side-stripper draws / vapour returns: control on CFU-100-PR-PID-008.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

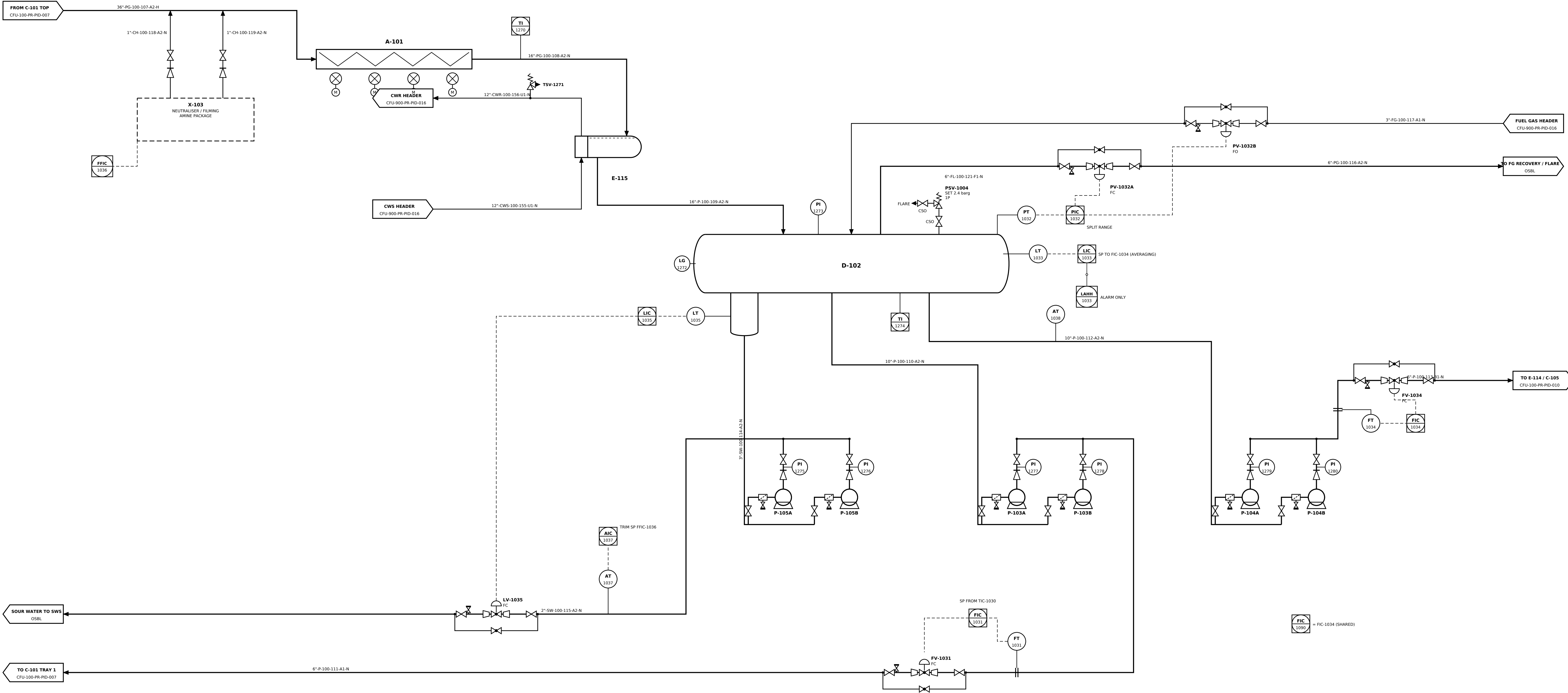
| REV                                                                                   | DATE       | DESCRIPTION              |            |
|---------------------------------------------------------------------------------------|------------|--------------------------|------------|
| A                                                                                     | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT           |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM<br>ATMOSPHERIC COLUMN C-101 - UPPER SECTION & PUMPAROUNDS |            |                          |            |
| DISCIPLINE                                                                            |            | STAGE                    | SHEET      |
| PROCESS                                                                               |            | FEED                     | 8 OF 17    |
| SCALE                                                                                 |            | NTS                      |            |
| DRAWN                                                                                 | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                                                           | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                                                                        |            |                          | REV        |
| CFU-100-PR-PID-007                                                                    |            |                          | A          |



- NOTES**
- Side-stripper vapours return to C-101 one tray above draw; strippers protected by PSV-1009 (fire case).
  - Stripping steam ratio to product rate (6 lb/bbl) via FIC-1054/1064/1074.
  - Product rundown temperatures: kerosene 45 C, diesel 55 C, AGO 60 C (air coolers with VFD on one fan).
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                                                           | DATE               | DESCRIPTION              |
|-------------------------------------------------------------------------------|--------------------|--------------------------|
| A                                                                             | 2026-10-02         | Issued for review (FEED) |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT   |                    |                          |
| PIPING & INSTRUMENT DIAGRAM<br>SIDE STRIPPERS C-102/103/104 & PRODUCT RUNDOWN |                    |                          |
| DISCIPLINE                                                                    | STAGE              | SCALE                    |
| PROCESS                                                                       | FEED               | NTS                      |
| DRAWN                                                                         | CHECKED            | APPROVED                 |
| CLAUDE CODE                                                                   | -                  | -                        |
| DRAWING NUMBER                                                                | CFU-100-PR-PID-008 |                          |
| REV                                                                           | A                  |                          |

|                                                                                                                                                                                                                       |                                                                                                                                                                                                             |                                                                                                                                                                              |                                                                                                                                            |                                                                                                                                                       |                                                                                                                                                   |                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>A-101</b><br>Atm. overhead condenser<br>1667 m2 bare; 3 bay(s) 6 m x 12 m, 6 fans<br>DESIGN 9.7 barg / 165 C<br>DUTY 33.75 MW; 6 x 55 kW fans<br>MAT'L CS tubes, Al fins (A-101: CS + inlet ferrules, Ti optional) | <b>E-115</b><br>Atm. overhead trim condenser<br>740 m2; 2 shell(s)<br>DESIGN 3.5 / 7 barg / 90 C; CA 3 mm<br>DUTY 5.96 MW<br>SHELL: OH vapour/condensate / TUBE: Cooling water<br>MAT'L CS shell / Ti tubes | <b>D-102</b><br>Atm. overhead reflux drum (3-phase)<br>ID 3.4 m x 12.0 m T/T; boot ID 1.5 m x 3.0 m<br>DESIGN 3.5 barg / 160 C; CA 6 mm<br>MAT'L CS (HIC-resistant), 6 mm CA | <b>P-103A/B</b><br>Atm. reflux<br>RATED 159 m3/h @ 102 m, dP 7 bar<br>MOTOR: 55 kW; API 682 Plan 11/52<br>DES. T 75 C; MAT'L S-6 (CS/L2Cr) | <b>P-104A/B</b><br>Unstabilised naphtha<br>RATED 180 m3/h @ 232 m, dP 16 bar<br>MOTOR: 160 kW; API 682 Plan 11/52<br>DES. T 75 C; MAT'L S-6 (CS/L2Cr) | <b>P-105A/B</b><br>Overhead sour water<br>RATED 14 m3/h @ 51 m, dP 2 bar<br>MOTOR: 5.5 kW; API 682 Plan 11/52<br>DES. T 75 C; MAT'L S-6 (CS/L2Cr) | <b>X-103</b><br>Neutraliser / filming amine package (atm OH)<br>Tank 2 m3 + 2 x 100 % metering pumps<br>MOTOR: 0.75 kW<br>MAT'L SS316 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|

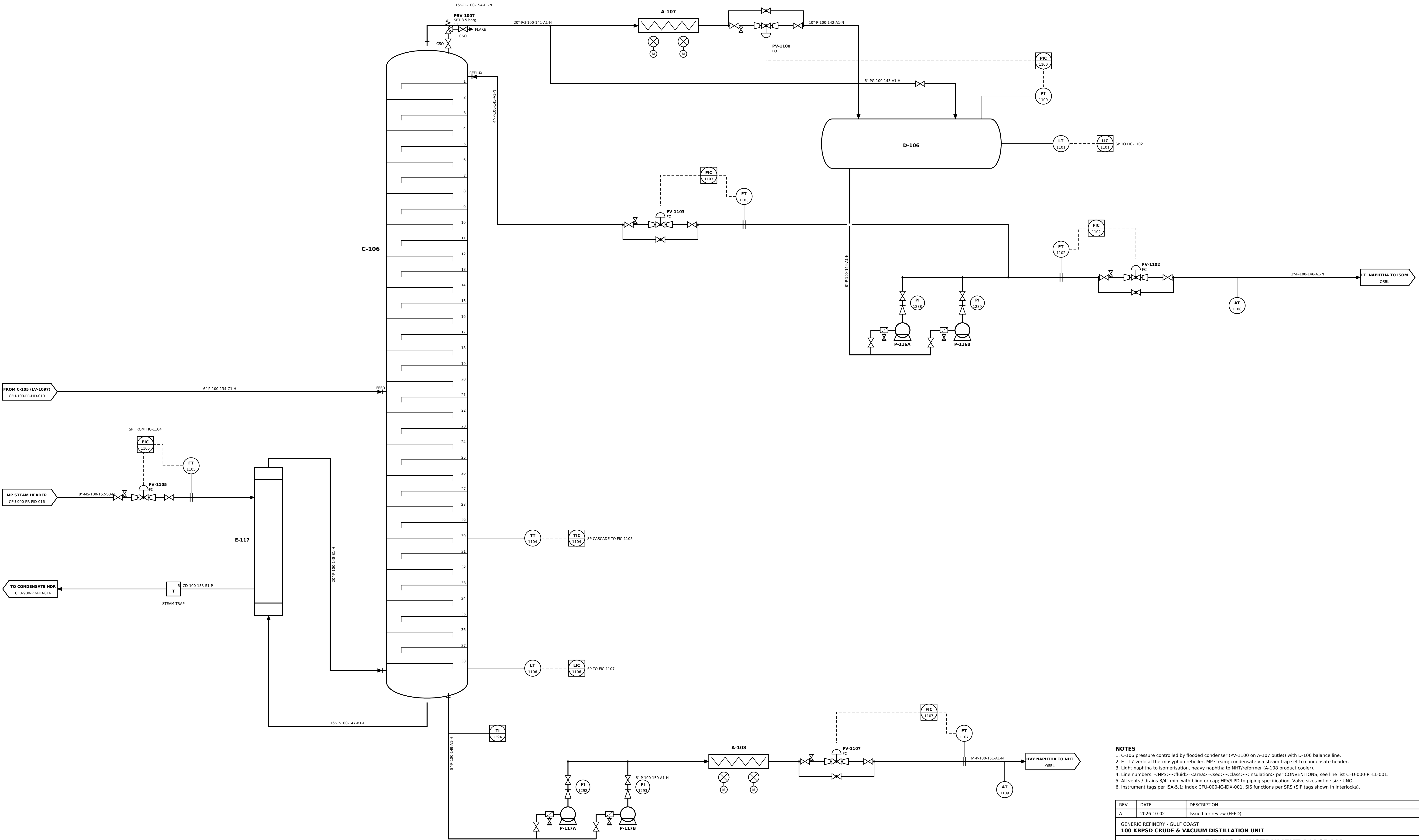


- NOTES**
- Neutraliser and filming amine injected upstream of A-101; pH control AIC-1037 trims FFIC-1036.
  - PIC-1032 split range: 0-50 % PV-1032B (FG make-up) closing, 50-100 % PV-1032A (off-gas) opening.
  - D-102 high-high level is a DCS alarm only (LAHH-1033); TSV on E-115 CW side, set per TSV-tyt.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                        | DATE       | DESCRIPTION              |            |
|--------------------------------------------|------------|--------------------------|------------|
| A                                          | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST              |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |            |
| C-101 OVERHEAD SYSTEM & REFLUX DRUM D-102  |            |                          |            |
| DISCIPLINE                                 | STAGE      | SCALE                    | SHEET      |
| PROCESS                                    | FEED       | NTS                      | 10 OF 17   |
| DRAWN                                      | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                             |            |                          | REV        |
| CFU-100-PR-PID-009                         |            |                          | A          |



|                                                                                                                                           |                                                                                                                                                                          |                                                                                                                                                                                                                         |                                                                                                              |                                                                                                                                                    |                                                                                                                                                     |                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C-106</b><br>Naphtha splitter<br>ID 3.1 m x 29.5 m TTT<br>DESIGN 3.5 barg / 185 C; CA 3 mm<br>38 valve trays, feed tray 21<br>MAT'L CS | <b>E-117</b><br>Splitter reboiler (MP steam)<br>225 m2; 1 shell(s)<br>DESIGN 5 / 13 barg / 280 C; CA 3 mm<br>DUTY 6.83 MW<br>SHELL: Naphtha / TUBE: MP steam<br>MAT'L CS | <b>A-107</b><br>Splitter overhead condenser<br>BB1 m2 bare; 2 bay(s) 6 m x 12 m, 4 fans<br>DESIGN 9.7 barg / 135 C<br>DUTY 9.63 MW; 4 x 30 kW fans<br>MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional) | <b>D-106</b><br>Splitter reflux drum<br>ID 2.7 m x 8.5 m TTT<br>DESIGN 3.5 barg / 120 C; CA 3 mm<br>MAT'L CS | <b>P-116A/B</b><br>Splitter reflux / LN<br>RATED 146 m3/h @ 111 m, dP 7 bar<br>MOTOR 55 kW, API 682 Flan 11/52<br>DES. T 80 C; MAT'L S-6 (CS/12Cr) | <b>P-117A/B</b><br>Heavy naphtha product<br>RATED 142 m3/h @ 98 m, dP 6 bar<br>MOTOR 45 kW, API 682 Flan 11/52<br>DES. T 185 C; MAT'L S-6 (CS/12Cr) | <b>A-108</b><br>Heavy naphtha product cooler<br>401 m2 bare; 1 bay(s) 6 m x 12 m, 2 fans<br>DESIGN 9.7 barg / 185 C<br>DUTY 5.81 MW; 2 x 30 kW fans<br>MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional) |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



- NOTES**
- C-106 pressure controlled by flooded condenser (PV-1100 on A-107 outlet) with D-106 balance line.
  - E-117 vertical thermosiphon reboiler, MP steam; condensate via steam trap set to condensate header.
  - Light naphtha to isomerisation, heavy naphtha to NHT/reformer (A-108 product cooler).
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                                                         | DATE       | DESCRIPTION              |            |
|-----------------------------------------------------------------------------|------------|--------------------------|------------|
| A                                                                           | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM<br>NAPHTHA SPLITTER C-106                       |            |                          |            |
| DISCIPLINE                                                                  | STAGE      | SCALE                    | SHEET      |
| PROCESS                                                                     | FEED       | NTS                      | 12 OF 17   |
| DRAWN                                                                       | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                                                 | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                                                              |            |                          | REV        |
| CFU-100-PR-PID-011                                                          |            |                          | A          |

**H-201**  
Vacuum heater  
14.7 MW abs. / 17.5 MW fired; 1-cell cabin, 4 passes, 44 rad. tubes 6" x 18.3 m  
DESIGN 19.8 barg / 580 C; CA 3 mm  
DUTY 14.69 MW  
MAT'L Tubes A335 P9 (9Cr-1Mo); convection P5 / CS studded

MP STEAM HEADER  
CFU-900-PR-PID-016

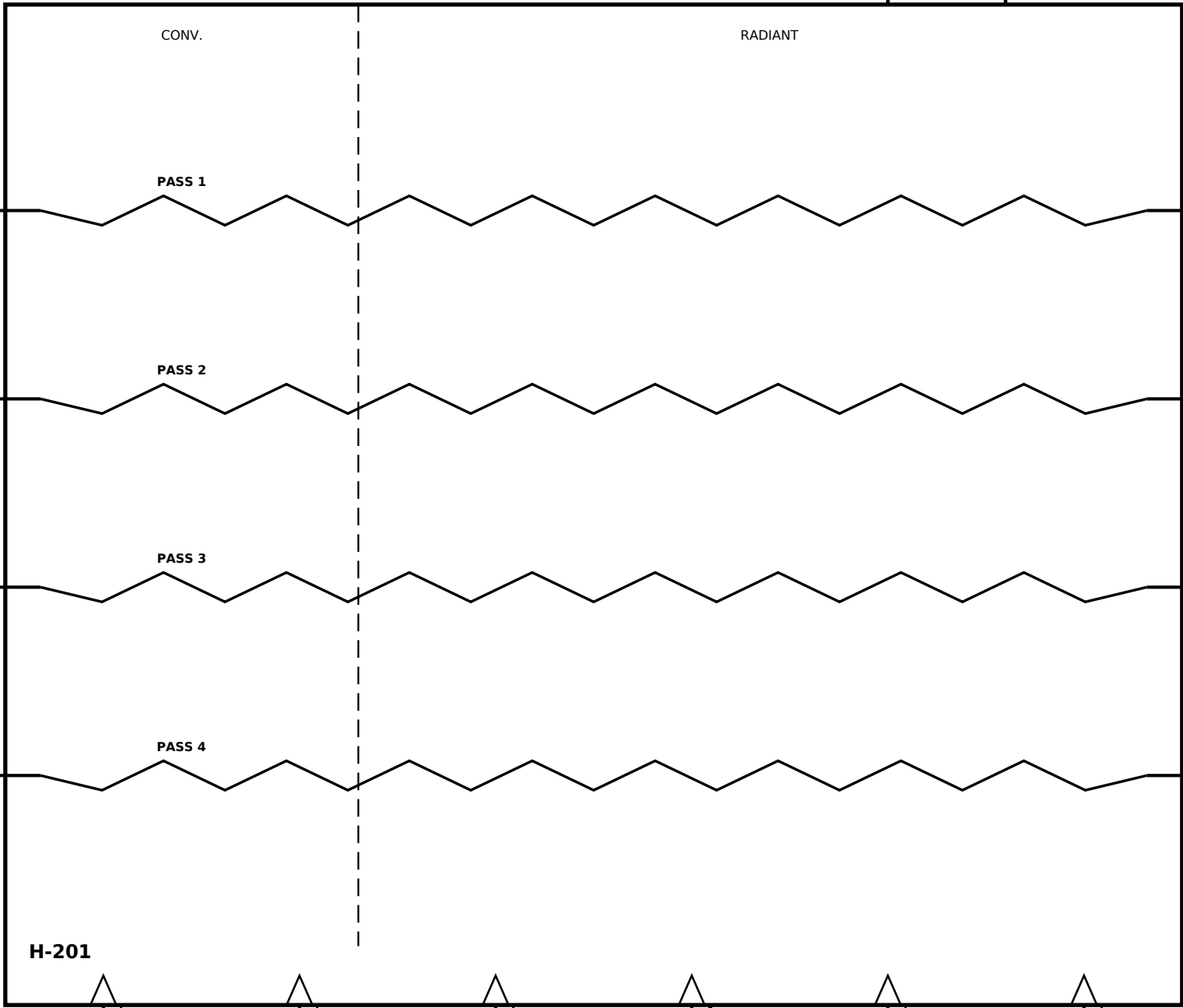
FROM P-112A/B (FV-1083)  
CFU-100-PR-PID-006

**H-201 BMS / SIS (SIL-RATED PLC)**

- SIF-201 (SIL 2)**  
H-201 low-low pass flow -> trip fuel
- SIF-202 (SIL 1)**  
H-201 low-low FG pressure / flame failure -> t

FUEL GAS FROM D-103  
CFU-900-PR-PID-016

PILOT GAS FROM D-103  
CFU-900-PR-PID-016



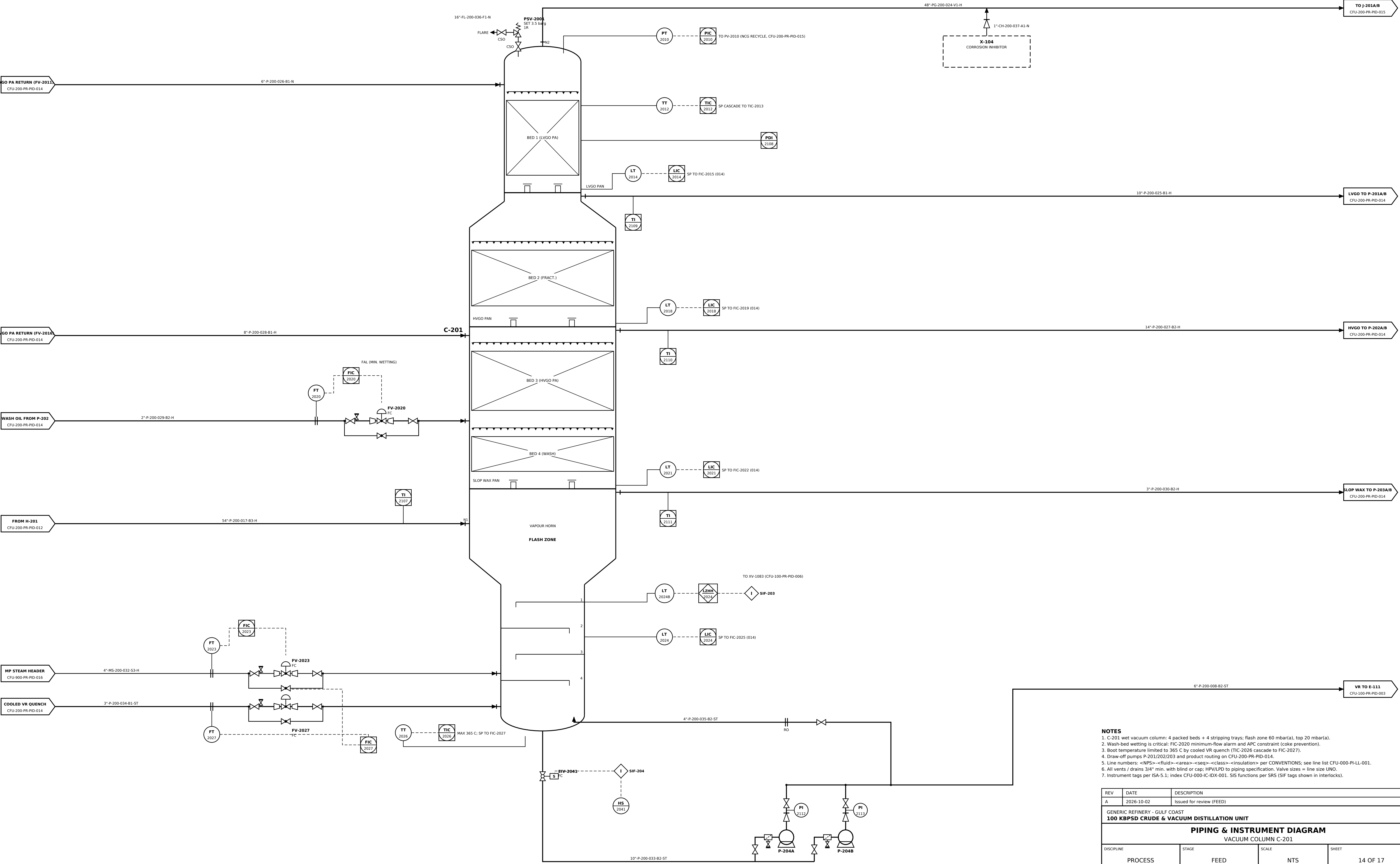
- NOTES**
- Passes shown schematically (convection + radiant in series); 4 identical passes, coil steam per pass.
  - Pass flow SIS transmitters FT-200xA/B/C (2003) independent of BPCS FT-200x (SIF-201).
  - Vacuum off-gas burned in dedicated H-201 burners via flame arrestor (from D-202 / PV-2031).
  - Natural draft heater: O2 trim AIC-2008 on stack damper.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                        | DATE       | DESCRIPTION              |            |
|--------------------------------------------|------------|--------------------------|------------|
| A                                          | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST              |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |            |
| VACUUM HEATER H-201                        |            |                          |            |
| DISCIPLINE                                 |            | STAGE                    | SCALE      |
| PROCESS                                    |            | FEED                     | NTS        |
|                                            |            | SHEET                    |            |
|                                            |            | 13 OF 17                 |            |
| DRAWN                                      | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                             |            |                          | REV        |
| CFU-200-PR-PID-012                         |            |                          | A          |

**C-201**  
Vacuum column (wet, packed)  
ID 4.4 m (top) / 6.6 m (main) / 3.0 m (boot) x 36.0 m T/T  
DESIGN PV: 3.5 barg / 415 °C, CA 6 mm  
4 packed beds (structured 250V/grid), 4 stripping trays, vapour horn  
MATL: CS + 410S clad (317L clad flash/wash zone)

**P-204A/B**  
Vacuum residue (incl. quench)  
RATED 196 m<sup>3</sup>/h @ 187 m, dP 14 bar  
MOTOR: 132 kW, API 682, Plan 538  
DES. T 395 °C, MATL: A-8 (316)

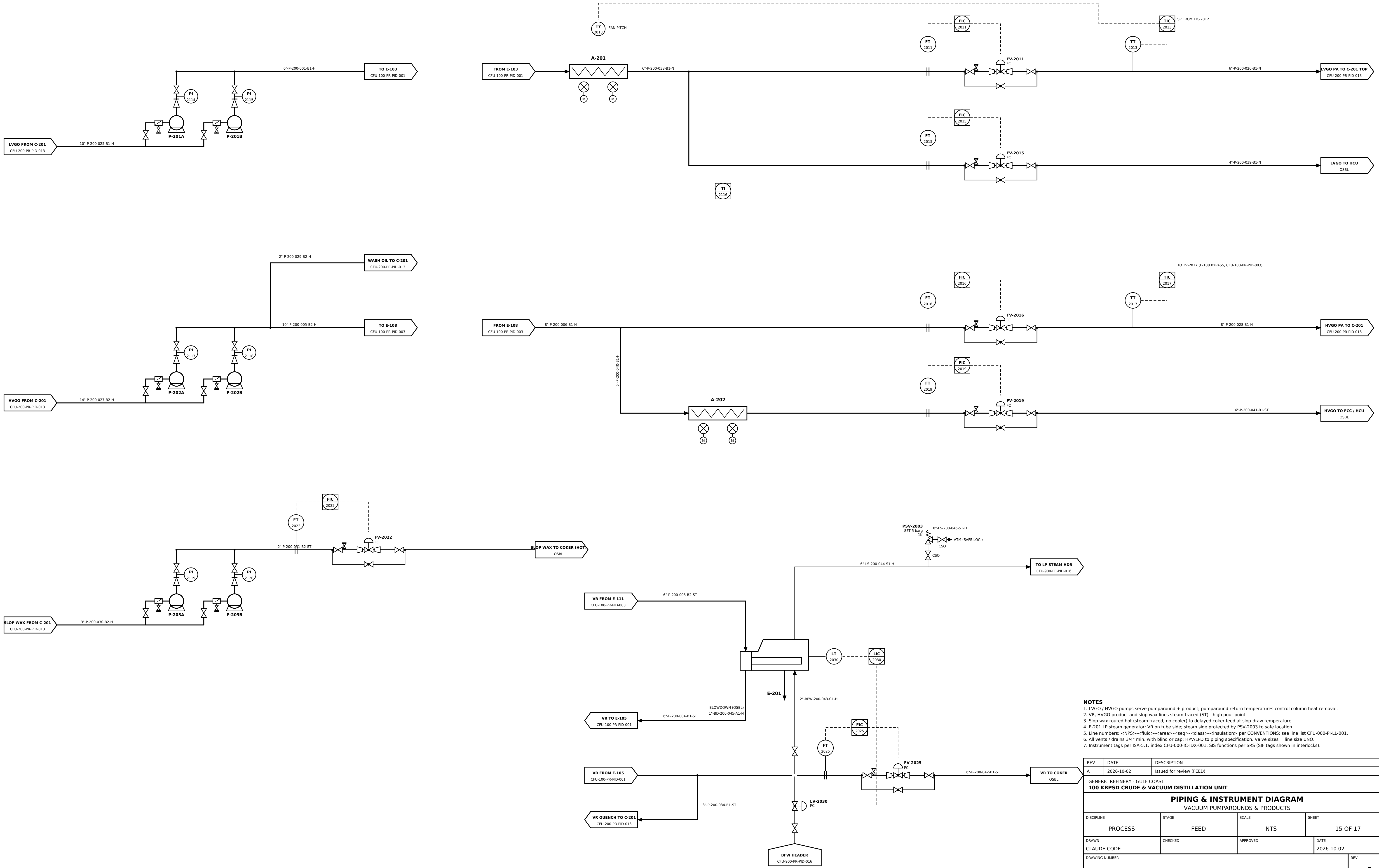
**X-104**  
Corrosion inhibitor package (VDU OH)  
Tank 2 m<sup>3</sup> + 2 x 100 % metering pumps  
MOTOR: 0.75 kW  
MATL: SS316

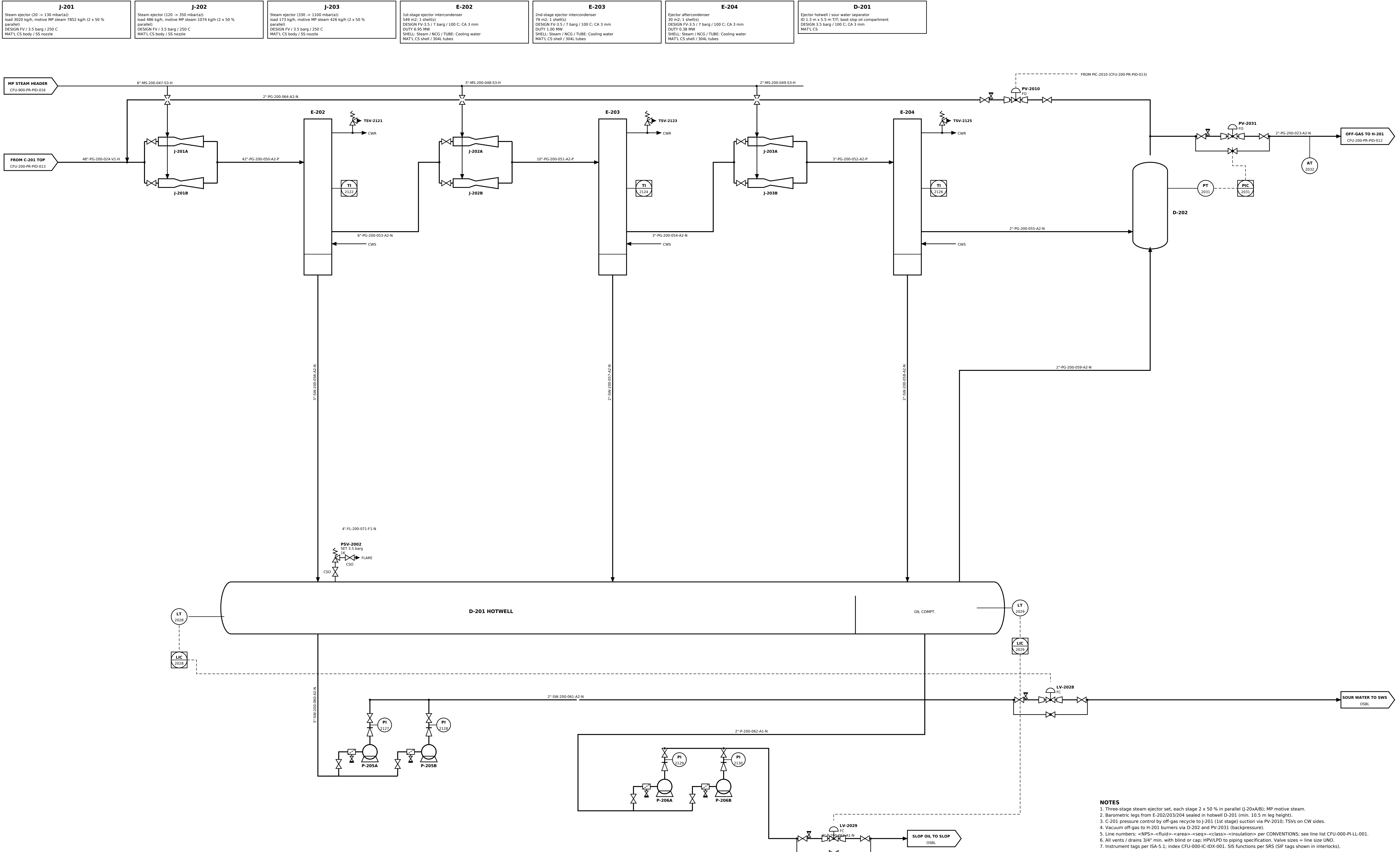


- NOTES**
- 1. C-201 wet vacuum column: 4 packed beds + 4 stripping trays; flash zone 60 mbar(a), top 20 mbar(a).
  - 2. Wash-bed wetting is critical: FIC-2020 minimum-flow alarm and APC constraint (coke prevention).
  - 3. Boot temperature limited to 365 °C by cooled VR quench (TIC-2026 cascade to FIC-2027).
  - 4. Draw-off pumps P-201/202/203 and product routing on CFU-200-PR-PID-014.
  - 5. Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - 6. All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - 7. Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                        | DATE       | DESCRIPTION              |
|--------------------------------------------|------------|--------------------------|
| A                                          | 2026-10-02 | Issued for review (FEED) |
| GENERIC REFINERY - GULF COAST              |            |                          |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |
| VACUUM COLUMN C-201                        |            |                          |
| DISCIPLINE                                 | STAGE      | SCALE                    |
| PROCESS                                    | FEED       | NTS                      |
| SHEET                                      |            | 14 OF 17                 |
| DRAWN                                      | CHECKED    | APPROVED                 |
| CLAUDE CODE                                | -          | -                        |
| DATE                                       |            | 2026-10-02               |
| DRAWING NUMBER                             |            | REV                      |
| CFU-200-PR-PID-013                         |            | A                        |

|                                                                                                                                                           |                                                                                                                                                      |                                                                                                                                   |                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P-201A/B</b><br>LVGO pumparound / product<br>RATED 198 m3/h @ 138 m, dP 10 bar<br>MOTOR 110 kW, API 682 Plan 1122<br>DES. T 255 C; MAT'L S-6 (CS/12Cr) | <b>P-202A/B</b><br>HVGO pumparound / product<br>RATED 391 m3/h @ 144 m, dP 10 bar<br>MOTOR 200 kW, API 682 Plan 538<br>DES. T 345 C; MAT'L A-8 (316) | <b>P-203A/B</b><br>Slop wax<br>RATED 12 m3/h @ 114 m, dP 8 bar<br>MOTOR 5.5 kW, API 682 Plan 538<br>DES. T 410 C; MAT'L A-8 (316) | <b>A-201</b><br>Lvgo cooler<br>397 m2 bare; 1 bay(s) 6 m x 12 m, 2 fans<br>DESIGN 9.7 barg / 125 C<br>DUTY 3.10 MW; 2 x 15 kW fans<br>MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional) | <b>A-202</b><br>Hvgo Product cooler<br>361 m2 bare; 1 bay(s) 6 m x 12 m, 2 fans<br>DESIGN 9.7 barg / 125 C<br>DUTY 6.64 MW; 2 x 37 kW fans<br>SHELL: BFW / LP steam / TUBE: Vacuum residue<br>MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional) | <b>E-201</b><br>VR / LP steam generator (kettle)<br>84 m2; 1 shell(s)<br>DESIGN 6 / 20 barg / 310 C; CA 3 mm<br>DUTY 1.78 MW<br>SHELL: BFW / LP steam / TUBE: Vacuum residue<br>MAT'L CS shell / Scr tubes |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|





COOLING WATER CONNECTIONS (HEADERS ON CFU-200-PR-PID-010):  
E-202: CWS 12" CWS-200-065-U1-N / CWR 12" CWR-200-066-U1-N  
E-203: CWS 6" CWS-200-067-U1-N / CWR 6" CWR-200-068-U1-N  
E-204: CWS 3" CWS-200-069-U1-N / CWR 3" CWR-200-070-U1-N

- NOTES**
- Three-stage steam ejector set, each stage 2 x 50 % in parallel (J-20xA/B); MP motive steam.
  - Barometric legs from E-202/203/204 sealed in hotwell D-201 (min. 10.5 m leg height).
  - C-201 pressure control by off-gas recycle to J-201 (1st stage) suction via PV-2010; TSVs on CW sides.
  - Vacuum off-gas to H-201 burners via D-202 and PV-2031 (backpressure).
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                        | DATE       | DESCRIPTION              |            |
|--------------------------------------------|------------|--------------------------|------------|
| A                                          | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST              |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |            |
| VACUUM EJECTOR SYSTEM & HOTWELL D-201      |            |                          |            |
| DISCIPLINE                                 |            | STAGE                    | SCALE      |
| PROCESS                                    |            | FEED                     | NTS        |
|                                            |            |                          | 16 OF 17   |
| DRAWN                                      |            | CHECKED                  | DATE       |
| CLAUDE CODE                                |            | -                        | 2026-10-02 |
| DRAWING NUMBER                             |            |                          | REV        |
| CFU-200-PR-PID-015                         |            |                          | A          |

D-103

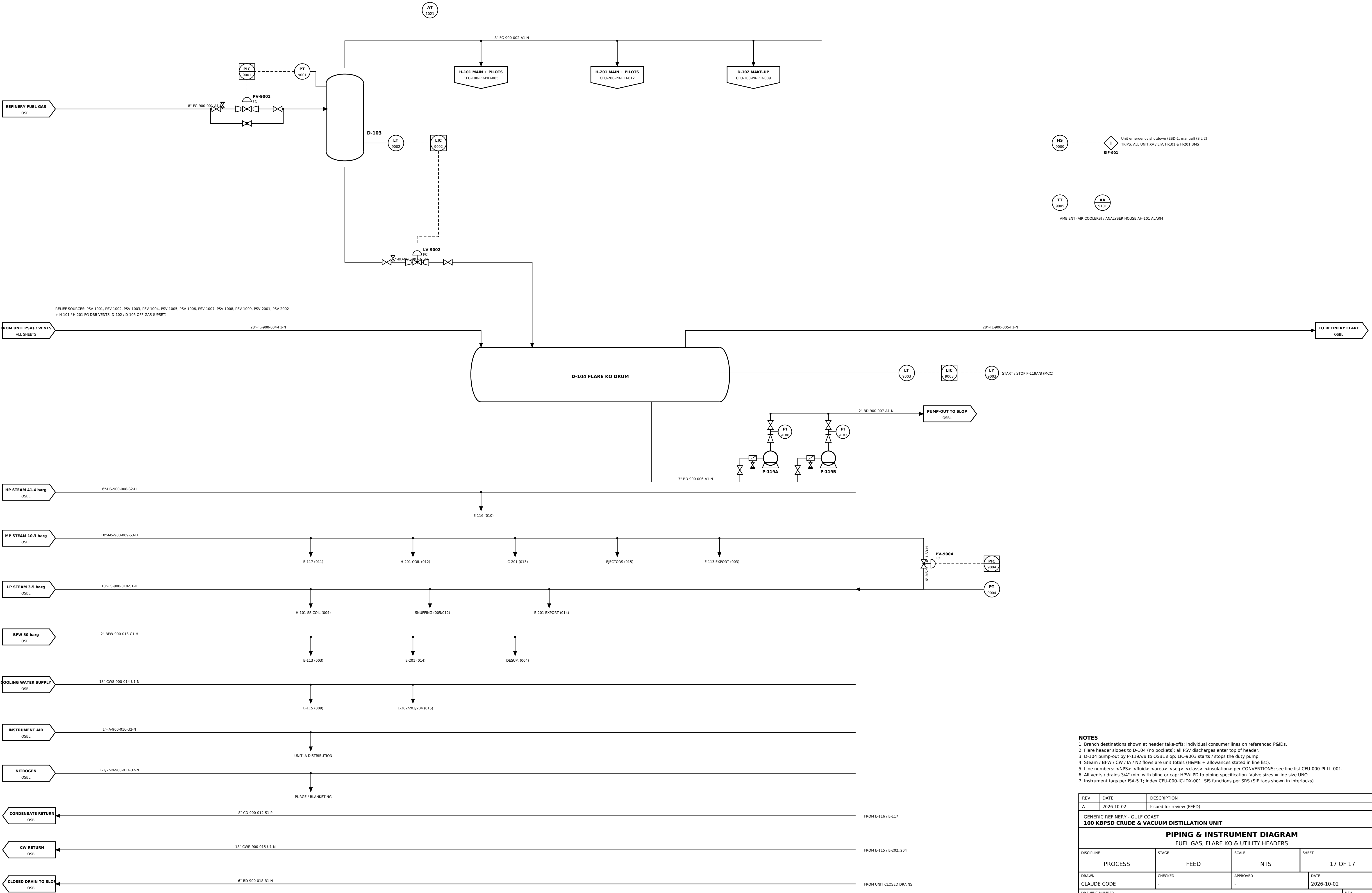
Fuel gas knock-out drum  
ID 0.9 m x 2.5 m TTT  
DESIGN 7.0 barg / 90 C; CA 3 mm  
MAT'L CS

D-104

Flare knock-out drum (unit)  
ID 2.1 m x 6.5 m TTT  
DESIGN 3.5 barg / 280 C; CA 3 mm  
MAT'L CS

P-119A/B

Flare KO drum pump-out  
RATED 21 m<sup>3</sup>/h @ 76 m, dP 6 bar  
MOTOR: 7.5 kW; API 682 Plan 11/92  
DES. T 90 C; MAT'L S-6 (CS/12Cr)



**NOTES**

- Branch destinations shown at header take-offs; individual consumer lines on referenced P&IDs.
- Flare header slopes to D-104 (no pockets); all PSV discharges enter top of header.
- D-104 pump-out by P-119A/B to OSBL slop; LIC 9003 starts / stops the duty pump.
- Steam / BFW / CW / IA / N<sub>2</sub> flows are unit totals (H6MB + allowances stated in line list).
- Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
- All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
- Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                                                         | DATE       | DESCRIPTION              |            |
|-----------------------------------------------------------------------------|------------|--------------------------|------------|
| A                                                                           | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM<br>FUEL GAS, FLARE KO & UTILITY HEADERS         |            |                          |            |
| DISCIPLINE                                                                  | STAGE      | SCALE                    | SHEET      |
| PROCESS                                                                     | FEED       | NTS                      | 17 OF 17   |
| DRAWN                                                                       | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                                                 | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                                                              |            |                          | REV        |
| CFU-900-PR-PID-016                                                          |            |                          | A          |